MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, LOW-POWER TYPE 2N1195

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a low-power, PNP, germanium, transistor.

1.2 Physical dimensions. See figure 1 (TO-5).

1.3 Maximum ratings.

P_T 1/ T_A = 25° C	I_E	I_C	V_{CBO}	V_{EBO}	V_{CEO}	T_{stg}
mW	mAdc	mAdc	Vdc	Vdc	Vdc	°C
250	40	-40	-30	-1.0	-20	-65 to +100

1/ Derate 3.33 mW/°C for $T_A > 25°$ C.

1.4 Primary electrical characteristics.

	$\lvert h_{fe} \rvert$ V_{CB} = -10 Vdc I_E = 10 mAdc f = 100 MHz	h_{fb} V_{CB} = -10 Vdc I_E = 10 mAdc	h_{ib} V_{CB} = -10 Vdc I_E = 10 mAdc	h_{ob} V_{CB} = -10 Vdc I_E = 10 mAdc
			ohms	μmho
Min	4	0.96	---	---
Max	---	0.995	10	20

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

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STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Transistor shall be of the design, construction, and physical dimensions shown on figure 1.

3.3.1 Lead material and finish. Lead material and finish shall be gold-plated Kovar. (Leads may be tin-coated if specified in the contract or order, see 6.2.)

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A, B, and C inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

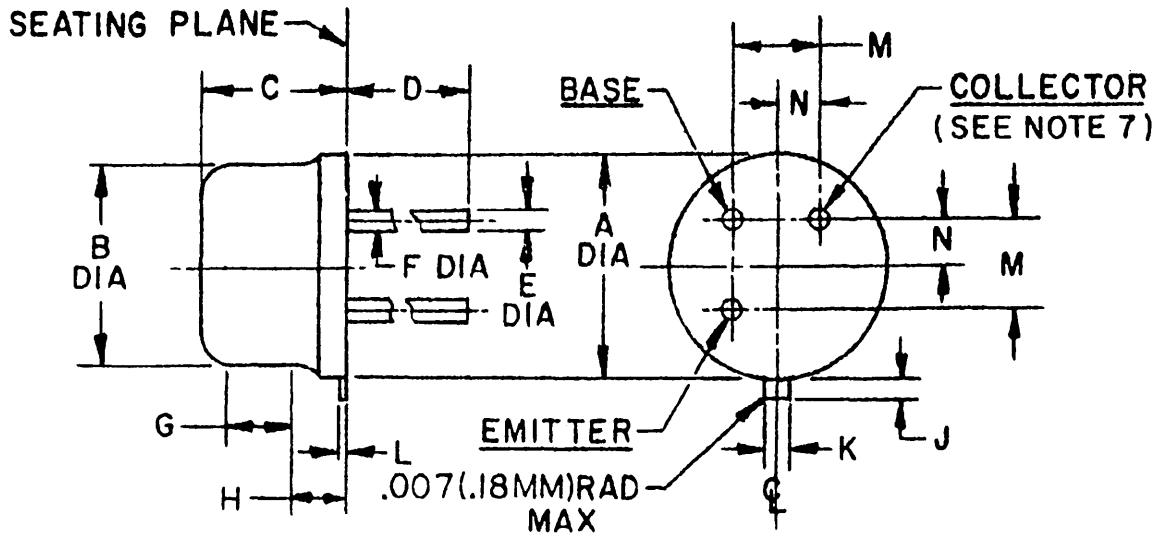
4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every 6 months during production.

4.3.4 Group B and group C life-test samples. Samples that have been subjected to group B, 340-hours life-test, may be continued on test for 1,000 hours in order to satisfy group C life-test requirements. These samples shall be predesignated, and shall remain subjected to the group C 1,000 hour acceptance evaluation after they have passed the group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1,000 hours shall be computed for 1,000-hour acceptance criteria, see 4.3.3.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III.

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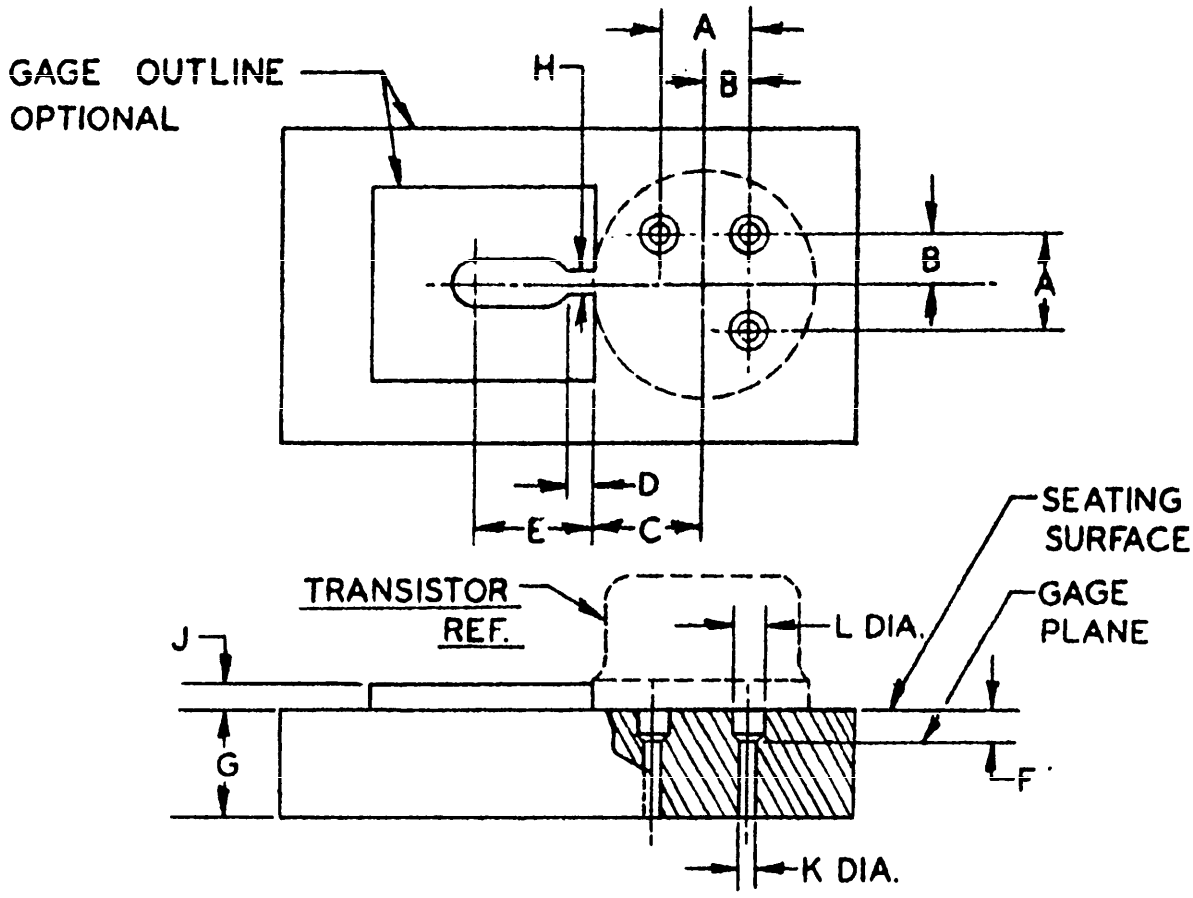


LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.335	.370	8.51	9.40	
B	.305	.335	7.75	8.51	
C	.240	.260	6.10	6.60	
D	1.500	1.750	38.10	44.45	9
E	.016	.021	.41	.53	2,9
F	.016	.019	.41	.48	3,9
G	.100	---	2.54	---	4
H	---	---	---	---	5
J	.029	.045	.74	1.14	8
K	.028	.034	.71	.86	
L	.009	.125	.23	3.18	
M	.1414 Nom		3.59 Nom		6
N	.0707 Nom		1.80 Nom		6

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. Variations on Dim B in this zone shall not exceed .010 (.25 mm).
5. Outline in this zone is not controlled.
6. When measured in a gaging plane .054+.001,-.000 (1.37+.03,-.00 mm) below the seating plane of the transistor max dia leads shall be within .007 (.18 mm) of their true location relative to a maximum width tab. Smaller dia leads shall fall within the outline of the max dia lead tolerance. Figure 2 preferred measured method.
7. The collector shall be internally connected to the case.
8. Measured from the maximum diameter of the actual device.
9. All 3 leads. (See 3.3.1).

FIGURE 1. Physical dimensions of transistor type 2N1195 (TO-5).

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NOTES:

1. The following gaging procedure shall be used: The use of a pin straightener prior to insertion in the gage is permissible. The device being measured shall be inserted until its seating plane is .125±.010 (3.18 mm ±.25 mm) from the seating surface of the gage. A spacer may be used to obtain the .125 (3.18 mm) distance from the gage seat prior to force application. A force of 8 oz ±.5 oz shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gage.
2. The location of the tab locator, within the limits of dim C, will be determined by the tab and flange dimension of the device being checked.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

DIMENSIONS				
LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	.1409	.1419	3.58	3.60
B	.0702	.0712	1.78	1.81
C	.182	.199	4.62	5.05
D	.009	.011	.23	.28
E	.125 Nom		3.18 Nom	
F	.054	.055	1.37	1.40
G	.372	.378	9.45	9.60
H	.0350	.0355	.89	.90
J	.150 Nom		3.81 Nom	
K	.0325	.0335	.83	.85
L	.0595	.0605	1.51	1.54

FIGURE 2. Gage for lead and tab location for transistor type 2N1195.

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TABLE I. Group A inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Breakdown voltage, collector to emitter	3011	Bias cond. D; I_C = -5.0 mAdc 1/		BV_{CEO}	-20	---	Vdc
Breakdown voltage, collector to base	3001	Bias cond. D; I_C = -100 μAdc		BV_{CBO}	-30	---	Vdc
Breakdown voltage, emitter to base	3026	Bias cond. D; I_E = -100 μAdc		BV_{EBO}	-1.0	---	Vdc
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -20 Vdc; I_E = 0		I_{CBO}	---	-5.0	μAdc
Emitter to base cutoff current	3061	Bias cond. D; V_{EB} = -1 Vdc; I_C = 0		I_{EBO}	---	-100	μAdc
Subgroup 3			5				
Small-signal short-circuit input impedance	3201	V_{CB} = -10 Vdc; I_E = 10 mAdc		h_{ib}	---	10	ohms
Small-signal open-circuit reverse-voltage transfer ratio	3211	V_{CB} = -10 Vdc; I_E = 10 mAdc		h_{rb}	---	$3x10^3$	---
Small-signal open-circuit output admittance	3216	V_{CB} = -10 Vdc; I_E = 10 mAdc		h_{ob}	---	20	μmho
Real part of small-signal short-circuit input impedance	3266	V_{CE} = -10 Vdc; I_E = 10 mAdc; f = 250 MHz		RE_{hie}	---	80	ohms
Open-circuit output capacitance	3236	V_{CB} = -10 Vdc; I_E = 0; 100 kHz $\leq$ f $\leq$ 1 MHz		C_{obo}	---	1.5	pf
Subgroup 4			5				
Small-signal short-circuit forward-current transfer ratio	3206	V_{CB} = -10 Vdc; I_E = 10 mAdc		h_{fb}	0.96	0.995	---
Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = -10 Vdc; I_C = 10 mAdc; f = 100 MHz		$\lvert h_{fe} \rvert$	4	---	---

1/ tp $\leq$ 100 msec, duty cycle $\leq$ 15%.

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TABLE II. Group B inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			20				
Physical dimensions	2066	(See figure 1)		---	---	---	---
Subgroup 2			15				
Solderability	2026	Omit aging		---	---	---	---
Thermal shock (temperature cycling)	1051	Test cond. B; except T(high) = +100°C		---	---	---	---
Thermal shock (glass strain)	1056	Test cond. A; T(high) = +85° ±5°C T(low) = 0° ±2°C		---	---	---	---
Seal (leak-rate)	---	MIL-STD-202, method 112, test cond. C, procedure III; test cond. B for gross leaks		---	---	$5x10^{-7}$	atm cc/sec
Moisture resistance	1021			---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -20 Vdc; I_E = 0		I_{CBO}	---	-5.0	μAdc
Small-signal short-circuit forward-current transfer ratio	3206	V_{CB} = -10 Vdc; I_E = 10 mAdc		h_{fb}	0.95	---	---
Subgroup 3			15				
Shock	2016	Nonoperating; 1,500 G, 0.5 msec, 5 blows in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
Vibration fatigue	2046	Nonoperating		---	---	---	---
Vibration, variable frequency	2056			---	---	---	---
Constant acceleration	2006	10,000 G; in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
End points: (Same as subgroup 2)							
Subgroup 4			20				
Terminal strength (lead fatigue)	2036	Test cond. E		---	---	---	---
Subgroup 5			20				
Salt atmosphere (corrosion)	1041			---	---	---	---
End points: (Same as subgroup 2)							

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TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 6			7				
High-temperature life (nonoperating)	1031	T_{stg} = +100°C; t = 340 hrs (see 4.3.4)		---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -20 Vdc		I_{CBO}	---	-10	µAdc
Small-signal short-circuit forward-current transfer ratio	3206	V_{CB} = -10 Vdc; I_E = 10 mAdc		h_{fb}	0.95	---	---
Subgroup 7			7				
Steady state operation life	1026	T_A = +25°C; P_T = 250 mW; I_C = 40 mAdc; t = 340 hrs (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6)							

TABLE III. Group C inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			20				
Thermal resistance	3151			θJ-A	---	0.3	°C/mW
Subgroup 2			λ=10				
High-temperature life (nonoperating)	1031	T_{stg} = +100°C (see 4.3.4)		---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -20 Vdc		I_{CBO}	---	-10	µAdc
Small-signal short-circuit forward-current transfer ratio	3206	V_{CB} = -10 Vdc I_E = 10 mAdc		h_{fb}	0.95	---	---
Subgroup 3			λ=10				
Steady state operation life	1026	T_A = +25°C; P_T = 250 mW; I_C = 40 mAdc (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 2)							

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5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, section 5.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data.

(a) Lead finish if other than gold-plated Kovar (see 3.3.1).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.4 Substitution criteria. The device covered herein is interchangeable with the device covered by the superseded MIL-S-19500/71C.

Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - EL, MU, MI
Navy - SH
Air Force - 11, 17, 85

User activities:
Army - SM
Navy - CG, MC, AS, OS
Air Force - 14, 19

Preparing activity:
Navy - SH

(Project 5961-0008-04)

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